John (Feng) Huang

Lead AI & Systems Engineer | Industrial Digital Transformation & Equipment Domain

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EDUCATION & KEY CREDENTIALS

Education: Beijing Jiaotong University (211 National Key) - B.S. in Materials Chemistry
Languages: English (Business Fluent, former on-site interpreter),
Certifications: PMP (Project Management Professional, PMI, 3A Top Score - 2024.07) | Anthropic AI Certified (Claude 101, Fluency, Agent Skills, Claude

EXECUTIVE SUMMARY

15+ years of dual-track engineering leadership bridging precision manufacturing (semiconductor/cleanroom FPD, MEMS medical devices, heavy machinery) and modern AI systems. Proven record from 0-to-1 photo line installation (overseas technical support in Japan & Turkey) to orchestrating group-wide EAM digitalization across 9 Business Units and 40+ factories, to independently architecting on-premise LLM/Agent microservices in EdTech. Deep engineering rigor in GMP validation (IQ/OQ/PQ) and SPC/DOE quality control, coupled with pragmatic AI engineering.

CORE COMPETENCIES & DOMAIN EXPERTISE

AI & Agent Engineering:	LLM Tool-Use / MCP Workflows, RAG Architecture, Local LLM Inference (vLLM / Qwen / bge-m3), LiteLLM Gateway, Agent Evaluation & Observability
Industrial & Quality Systems:	EAM & MES Architecture, Time-Series Industrial Data Governance, GMP Validation (IQ/OQ/PQ), SPC / CPK Analysis (1.67), DOE, 8D Root Cause, SAP Integration
Platform & Software:	Python (FastAPI, Asyncio, Data Pipelines), Linux System Engineering & Automation, Docker, SQL / PostgreSQL (+pgvector), Pragmatic Vibe Coding
Cross-Border Execution:	Autonomous Execution, Asynchronous Remote Delivery, Technical Documentation, Proven On-Site International Tech Support (Turkey & Japan)

PROFESSIONAL EXPERIENCE

AI Product Manager / Lead Systems Architect
2026.05 - 2026.09

Xinghuo Shuzhi (K12 AI EdTech)

- Architected pluggable AI foundation (agent-base) and delivered 3 production microservices: question intelligence (dual knowledge graphs + VLM), explainable student mastery algorithms, and personalized tutoring generation.
- Deployed 100% on-premise model matrix (Qwen3.6-35B / Qwen3.8-27B / bge-m3) with LiteLLM gateway, strictly ensuring sensitive education data never left local intranet.
- Built document AI platform (X-AnyLabeling-Web) from 0 to 1: workbook segmentation, RapidOCR, KaTeX formula editing, and YOLOv8 pre-labeling with 3-tier QC workflow (FastAPI + Vue 3).

Digital Equipment Management Manager / EAM Transformation Lead
2021.06 - 2025.11

Sany Group (Heavy Construction Machinery)

- Led enterprise EAM rollout across 9 Business Units and 40+ factories, achieving 100% online operational coverage and closing data loop between SAP depreciation and MES OEE.
- Pioneered on-premise LLM+RAG Maintenance AI Assistant (Dify + Qwen/Rerank), reaching 63% solution adoption rate and drastically mitigating senior technician shortages.
- Standardized equipment master data across 300K+ records (+40% cleanliness); built multi-criteria evaluation model redeploying ~$3M USD in idle assets; reduced annual cash equipment expenses by ~8%.

Lead MEMS Equipment Engineer
2018.09 - 2021.06

Weizhi Medical Devices (Precision Cleanroom Startup)

- Sole equipment lead: established complete maintenance, SOP, and spare-parts systems from scratch for 27 cleanroom tools (photolithography, coating, etching, thin-film).
- Mastered GMP validation standards: independently completed IQ/OQ/PQ 3-stage qualifications for high-precision Suss aligner and MicroWriter ML3, achieving 100% regulatory audit compliance.

Senior Coating Process & Yield Improvement Engineer
2010.07 - 2018.08

Tianma Microelectronics & CSOT (China Star Optoelectronics)

- Led photolithography in-line tool ramp-up from 0-2K to 30K/mo (completed vendor training & tool sign-off in Japan); boosted equipment uptime from 75% to 95% and cut takt time (65s to 53s).
- Targeted critical yield bottlenecks: resolved coating defects (15 to 1/mo) and reduced rework rate from 2817 to 135 pcs/mo; led OLED lifetime enhancement project via DOE (lifetime +35%).
- Maintained strict SPC quality control (CPK 1.33+ / 1.67 for high-end accounts); served as on-site technical interpreter in Turkey and internal TPM technical trainer.